

Introduction to Social Data Analytics: Week 8 Part 1

Today's Lecture

- ▶ Introduce the empirical application of the week: Greenstone et al. (2022)
- ▶ Discuss how to make an assortment of figures in base R
 - ▶ histograms: how does pollution vary across cities in China?
 - ▶ box plots: what is a boxplot and how can we use it to study changes in pollution over time?

Application: Air Pollution Monitoring

- ▶ Greenstone et al. (2022) examine the introduction of automatic air pollution monitoring in China
 - ▶ Local officials were given incentives to meet pollution goals
 - ▶ But they also had the potential to manipulate local pollution data
- ▶ Greenstone et al. (2022) study how reported levels of PM10 change after automated monitoring of pollution levels
- ▶ PM10 is a measure of particulates in the air that are 10 micrometers in diameter or smaller

Application: Air Pollution Monitoring

- ▶ In R, we can load Stata datasets in a couple of ways
- ▶ One way is through the haven package
- ▶ Load data:

```
library(haven)  
library(tidyverse)
```

```
## Warning: package 'tibble' was built under R version 4.1.2
```

```
pm <- read_dta("station_day_1116.dta")  
pm <- filter(pm, !is.na(pm10))
```

- ▶ The final line filters out observations with missing pollution data

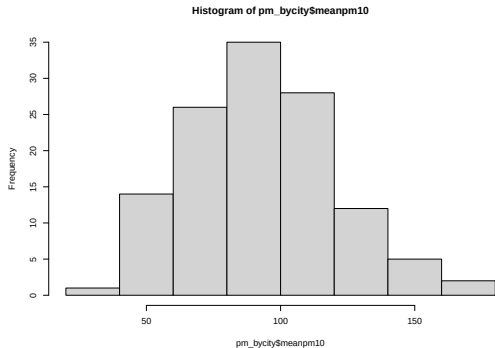
Group by city:

```
#Get average pm10 by city
pm_bycity <- pm %>%
  group_by(code_city) %>%
  summarize(meanpm10 = mean(pm10, na.rm=T))
#Look at the data
head(pm_bycity)
```

```
## # A tibble: 6 x 2
##   code_city meanpm10
##   <dbl>     <dbl>
## 1   110100     110.
## 2   120100     122.
## 3   130100     166.
## 4   130200     132.
## 5   130300      92.2
## 6   130400     145.
```

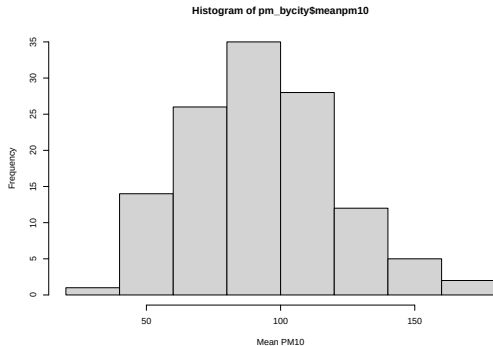
Create a basic histogram

```
hist(pm_bycity$meanpm10)
```



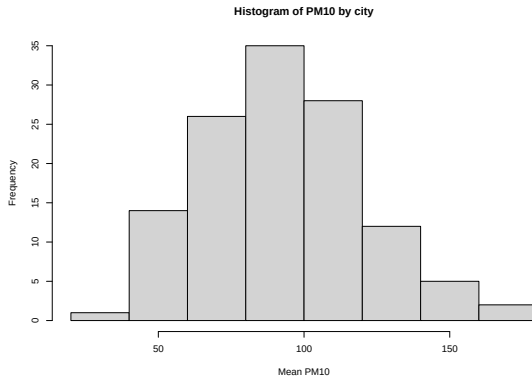
Add x and y axis labels

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",  
     ylab="Frequency")
```



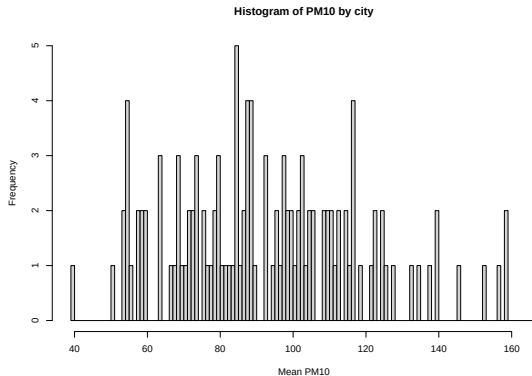
Add title

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",  
     ylab="Frequency",  
     main="Histogram of PM10 by city")
```



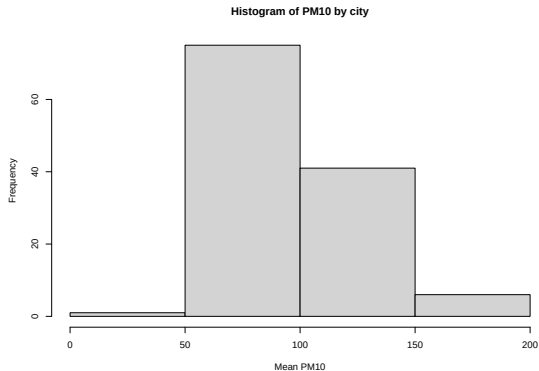
More breaks

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",  
     ylab="Frequency",  
     main="Histogram of PM10 by city", breaks=100)
```



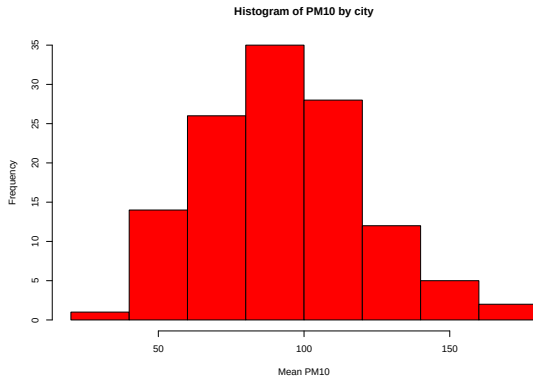
Fewer breaks

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",  
     ylab="Frequency",  
     main="Histogram of PM10 by city", breaks=4)
```



Add color

```
hist(pm_bycity$meanpm10, xlab="Mean PM10",  
     ylab="Frequency",  
     main="Histogram of PM10 by city", col="red")
```



► For more colors in R: <http://www.stat.columbia.edu/~tzheng/files/Rcolor.pdf>

Comparing Histograms

- ▶ As part of our analysis, we are going to compare two histograms, one before and one after automation

Split data into before and after automation

#Create data for before automation

```
pm_bycitybefore <- pm %>%  
  mutate(T=date - auto_date) %>%  
  filter(T< 0) %>%  
  group_by(code_city) %>%  
  summarize(meanpm10 = mean(pm10))
```

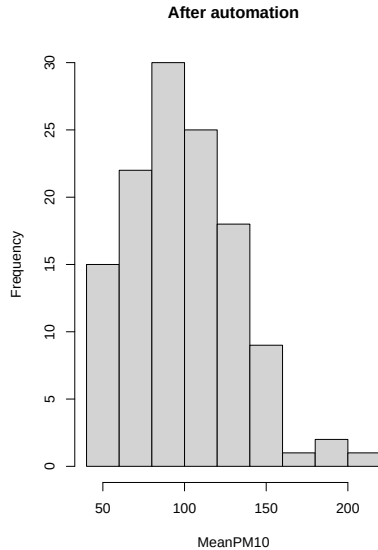
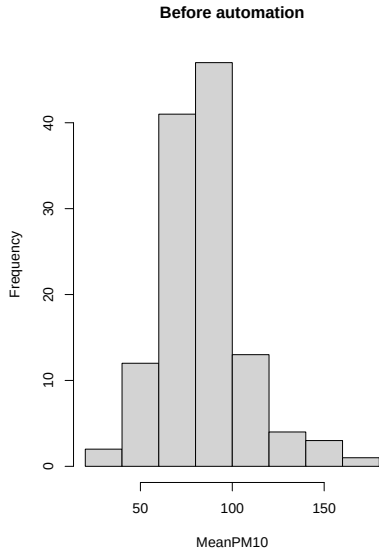
#Create data for after automation

```
pm_bycityafter <- pm %>%  
  mutate(T=date-auto_date) %>%  
  filter(T > 0) %>%  
  group_by(code_city) %>%  
  summarize(meanpm10 = mean(pm10))
```

Create two plots

```
#Create two panes for plots
par(mfrow=c(1,2))
#Plot histogram for before
hist(pm_bycitybefore$meanpm10, xlab="MeanPM10",
     ylab="Frequency",
     main="Before automation")
#Plot histogram for after
hist(pm_bycityafter$meanpm10, xlab="MeanPM10",
     ylab="Frequency",
     main="After automation")
```

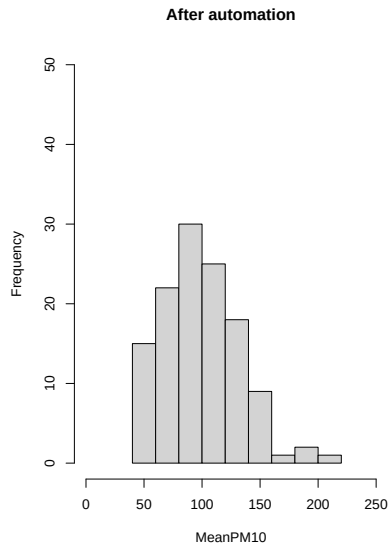
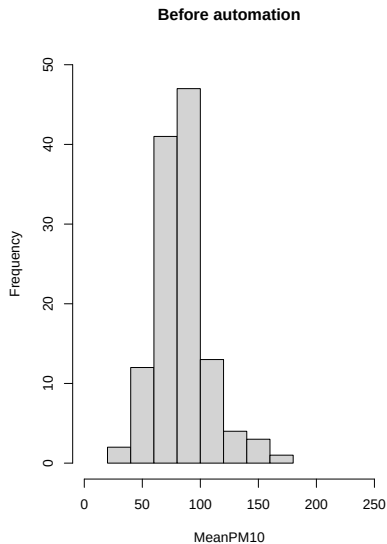
Create two plots



Change x- and y-axis limit to make them comparable

```
#Create two panes for plots
par(mfrow=c(1,2))
#Plot histogram for before
hist(pm_bycitybefore$meanpm10, xlab="MeanPM10",
     ylab="Frequency",
     main="Before automation",
     xlim=c(0,250), ylim=c(0,50))
#Plot histogram for after
hist(pm_bycityafter$meanpm10, xlab="MeanPM10",
     ylab="Frequency",
     main="After automation",
     xlim=c(0,250), ylim=c(0,50))
```

Change x- and y-axis limit to make them comparable



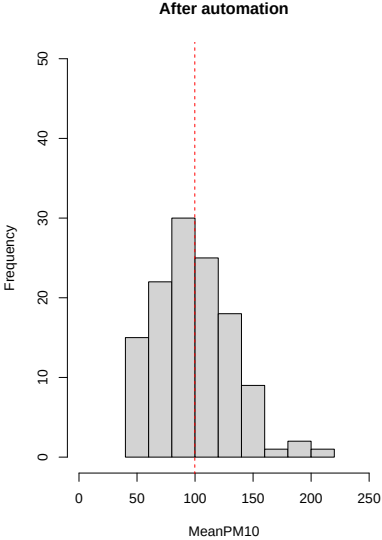
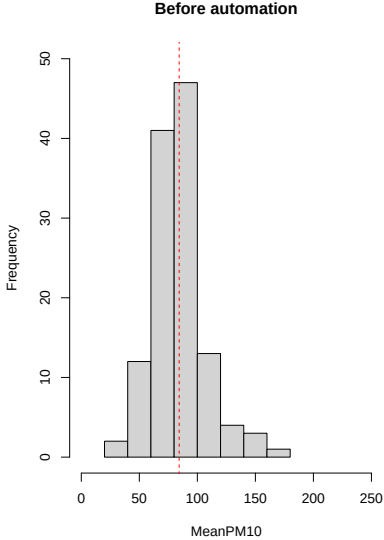
Adding lines to a plot

- ▶ We can use the function `lines()` to add a line to the plot
- ▶ To do so we must specify (x_1, x_2) and (y_1, y_2) . The line starts at the point (x_1, y_1) and ends at (x_2, y_2) .
- ▶ In our graph, if we want to place vertical line at the mean we can specify `lines(c(mean,mean),c(-10,100))`, where `mean` is an object that stores the mean value of pollution
 - ▶ In this example the vertical axis goes from 0 to 50. By specifying `c(-10,100)` the line will span the entire graph.

Add lines to the plot

```
meanbefore <- mean(pm_bycitybefore$meanpm10)
meanafter <- mean(pm_bycityafter$meanpm10)
#Create two panes for plots
par(mfrow=c(1,2))
#Plot histogram for before
hist(pm_bycitybefore$meanpm10, xlab="MeanPM10",
     ylab="Frequency",
     main="Before automation",
     xlim=c(0,250), ylim=c(0,50))
lines(c(meanbefore, meanbefore), c(-10, 100),
     lty=2, col="red")
#Plot histogram for after
hist(pm_bycityafter$meanpm10, xlab="MeanPM10",
     ylab="Frequency",
     main="After automation",
     xlim=c(0,250), ylim=c(0,50))
lines(c(meanafter, meaafter), c(-10, 100),
     lty=2, col="red")
```

Add lines to the plot



Boxplot

- ▶ A box plot is a figure we have not plotted yet
 - ▶ It displays a few relevant statistics about a variable of interest, such as the median, 25th percentile, and 75th percentile
 - ▶ It is useful when you want to compare these statistics across groups
- ▶ In our case, we want to understand the difference in these statistics before and after automation

Group air pollution by time to automation

```
#Group by day  
pm_byday <- pm %>%  
  mutate(  
    T=date-auto_date) %>%  
  filter(T>-364, T<364) %>%  
  group_by(T) %>%  
  summarize(meanpm10 = mean(pm10))
```

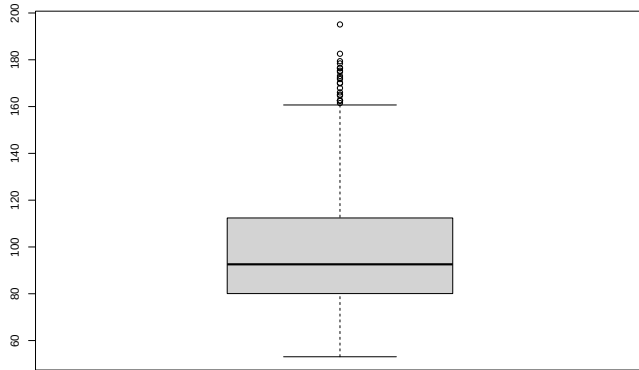
Make a boxplot for mean PM10

- ▶ Middle line of boxplot is median
- ▶ Upper and lower of box are third and first quartile
- ▶ Inter-quartile range (IQR): length of box
- ▶ Upper whisker: Add $1.5 * IQR$, or maximum
- ▶ Lower whisker: Add $1.5 * IQR$, or minimum

#Create Boxplot

```
boxplot(pm_byday$meanpm10)
```

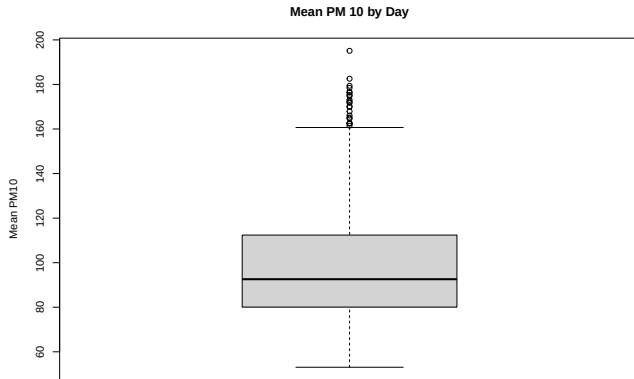
Make a boxplot for mean PM10



Customize the boxplot

```
#Create Boxplot
```

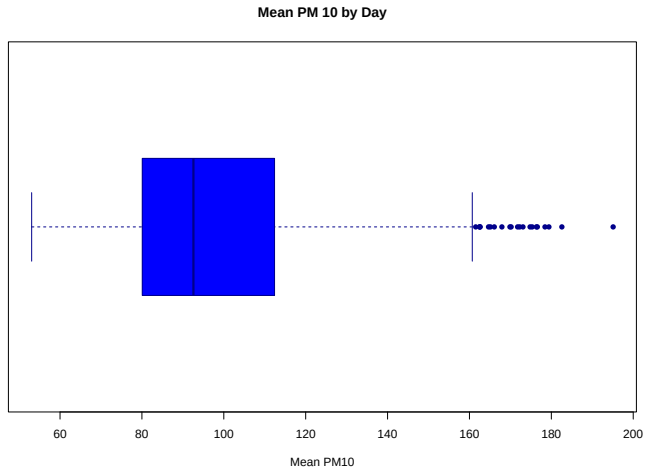
```
boxplot(pm_byday$meanpm10, ylab="Mean PM10",  
        main="Mean PM 10 by Day")
```



Customize the boxplot

```
#Change color, flip horizontal  
boxplot(pm_byday$meanpm10, xlab="Mean PM10",  
main="Mean PM 10 by Day",  
col="blue", border="darkblue",  
pch=16, horizontal=T)
```

Customize the boxplot



Plot by another variable

```
#Create after variable  
pm_byday$after <- pm_byday$T>0  
#Split into two boxplots  
boxplot(pm_byday$meanpm10 ~ pm_byday$after, xlab="Mean PM10",  
        main="Mean PM 10 by city", col="blue", border="darkblue",  
        pch=16, horizontal=T, ylab="After Automation")
```

Plot by another variable

